

Lecture 3: Prospect Theory

Decision Under Risk — Beyond Expected Utility

Prof. Muhebullah Karimzada

School of Economics, University of Northern British Columbia

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Opening Game — Your Risk Preferences

Make your choices — write them down

Problem 1: A) Win \$900 for certain. B) 90% chance of \$1,000, 10% chance of \$0.

Problem 2: A) Lose \$900 for certain. B) 90% chance of losing \$1,000, 10% chance of losing \$0.

Problem 3: You have \$2,000. A) Lose \$500 for certain. B) 50% chance of losing \$1,000.

Typical Results

- Problem 1: 84% choose A — risk averse in *gains*
- Problem 2: 87% choose B — risk *seeking* in *losses*
- Problem 3: 69% choose B — frame as loss $\Rightarrow$ gamble to avoid certain loss

EU cannot explain this pattern. Prospect Theory can.

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1. Why EU fails — the four-fold pattern
2. Reference dependence — gains and losses, not final wealth
3. Loss aversion — $\lambda \approx 2.25$
4. The S-shaped value function — THE key diagram
5. Probability weighting — the inverse-S function
6. Cumulative Prospect Theory (Tversky & Kahneman, 1992)
7. Applications: disposition effect, endowment effect, insurance, housing, health

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The Four-Fold Pattern

	Gains	Losses
High Probability	Risk Averse Prefer certain gain <i>"Lock in the win"</i>	Risk Seeking Gamble to avoid certain loss <i>"Roll the dice"</i>
Low Probability	Risk Seeking Buy lottery tickets <i>"Dream big"</i>	Risk Averse Buy insurance <i>"Protect against disaster"</i>

EU Prediction

With a concave utility function, EU predicts *always* risk averse — for all probabilities and gains/losses. The four-fold pattern is a systematic empirical refutation.

Framing Effects Cannot Be Explained by EU

The Asian Disease Problem (Kahneman & Tversky, 1981)

A disease threatens 600 lives. Two programmes available:

Gain Frame

- A: 200 people saved
- B: 1/3 prob 600 saved, 2/3 prob 0 saved

72% choose A (safe option)

Loss Frame

- C: 400 people die
- D: 1/3 prob 0 die, 2/3 prob 600 die

78% choose D (risky option)

$A = C$ and $B = D$. EU Cannot Explain This.

Under EU, utility is over final outcomes (lives saved). Describing the same outcome as “lives saved” vs “deaths” should not matter. But it does — dramatically. We need a theory with a **reference point**.

Core Idea of Prospect Theory

People evaluate outcomes as **gains or losses** relative to a **reference point**, not as levels of final wealth. The same final wealth can be a gain or a loss depending on the reference.

What Determines the Reference Point?

- **Status quo:** Current situation (most common)
- **Expectations:** What you expected to happen
- **Social comparison:** What peers received
- **Budget/target:** What you planned to spend

Reference Point Malleability

Getting a \$500 raise feels like:

- A **GAIN** if you expected \$500
- A **LOSS** if you expected \$1,000

Marketers exploit this: “Compare at \$200 — our price: \$150.” The \$200 is the reference point, making \$150 feel like a \$50 gain.

Nominal Wage Rigidity (Akerlof, Dickens & Perry, 1996)

Workers resist **nominal wage cuts** even when real wages are falling due to inflation. A 2% nominal raise with 5% inflation (-3% real) is *accepted*. A 3% nominal cut with 0% inflation (-3% real) is *resisted*. The reference point is the *nominal* wage — not the real value.

Shafir, Diamond & Tversky (1997)

Workers preferred a **5% raise** (with 12% inflation = -7% real) over a **2% raise** (with 4% inflation = -2% real). The preferred scenario left them *worse off in real terms* — because they evaluated the nominal gain, not the real outcome.

Canadian Wage Data

During the 1970s inflation spike (1974–1980, average 9%), Canadian nominal wages rose 10–12% annually — preserving the *perception* of real gains while real wages stagnated.

Loss Aversion — The Central Finding

Tversky & Kahneman (1992): $\lambda \approx 2.25$ — losses loom $2.25\times$ larger than equivalent gains.

- 50/50 bet — win $\$X$ or lose $\$100$: most require $X > \$200$.
- Butter consumer panel: $\lambda \approx 2.4$ (Hardie et al., 1993).
- Capuchin monkeys show loss aversion! (Chen et al., 2009).
- Workers accept lower wages when risk framed as bonuses vs penalties (Fryer et al., 2012).

Operates in: stock markets (disposition effect), labour (wage rigidity), housing (nominal loss refusal), health decisions, every negotiation.

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The S-Shaped Value Function — Properties

Kahneman & Tversky (1979, 1992)

The value function $v(x)$ has three key properties:

1. **Reference dependence:** Defined over *changes* x from a reference point r , not final wealth.
 $v(0) = 0$.
2. **Concave in gains** ($x > 0$): Risk aversion for gains. *“The first \$100 gained feels great; the next \$100 less so.”*
3. **Convex in losses** ($x < 0$): Risk seeking for losses. *“The first \$100 lost hurts most; subsequent losses hurt less — we adapt.”*
4. **Steeper for losses:** $|v(-x)| > v(x)$ for all $x > 0$. Loss aversion: $\lambda = v'(0^-)/v'(0^+) \approx 2.25$.

Parametric Form (Tversky & Kahneman, 1992)

$$v(x) = \begin{cases} x^\alpha & \text{if } x \geq 0 \\ -\lambda(-x)^\beta & \text{if } x < 0 \end{cases} \quad \text{with } \alpha = \beta = 0.88, \lambda = 2.25.$$

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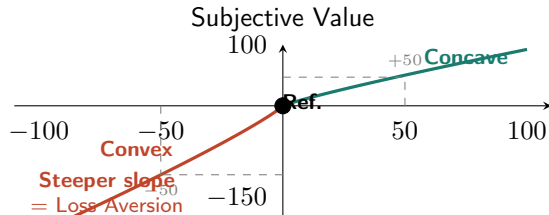
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THE Diagram — The S-Shaped Value Function



Reading the Diagram

- Green curve (gains): concave — risk averse
- Red curve (losses): convex — risk seeking
- Curves cross at the **reference point** (0,0)
- The red curve is **steeper** — loss aversion $\lambda \approx 2.25$

Key Numbers

$v(+50) \approx 47$ (gain); $v(-50) \approx -113$ (loss); ratio ≈ 2.4 . Losing \$50 hurts more than twice as much as gaining \$50.

The Endowment Effect (Kahneman, Knetsch & Thaler, 1990)

Giving up something you own is a *loss*; acquiring it would be a *gain*. Since losses loom larger: Willingness to Accept (WTA) $\approx 2 \times$ Willingness to Pay (WTP) for the **same object**.

The Mug Experiment

- Half of students given a coffee mug.
- **Sellers** (mug owners): median WTA = \$7.12
- **Buyers** (no mug): median WTP = \$2.87
- *Same mug — price gap driven purely by ownership.*

Economic Consequences

- Housing: homeowners demand too much to sell.
- Stocks: reluctance to sell holdings (disposition effect).
- Policy: losses of welfare opposed more than equivalent gains are welcomed.
- Jobs: laid-off workers resist equivalent re-employment.

How We Weight Probabilities

People **overweight** small probabilities (lottery, insurance) and **underweight** moderate-to-large ones. EU's linear treatment is wrong.

The Certainty Effect (Kahneman & Tversky, 1979)

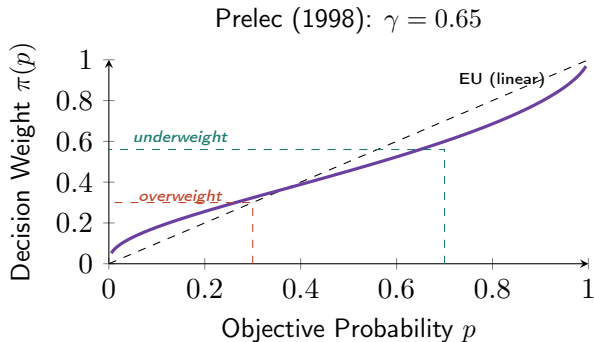
- Prefer \$2,400 certain over \$2,500@33% + \$0@67%.
- BUT prefer \$2,500@33% over \$2,400@34%.

Dropping from 100% → 99% *feels much larger* than 34% → 33%.

Psychological Explanation

- **Certainty effect:** 100% is qualitatively different from “almost certain.”
- **Possibility effect:** Non-zero is qualitatively different from impossible.
- These are *emotions about probabilities*.

The Probability Weighting Function



Properties: $\pi(0) = 0$, $\pi(1) = 1$.

Overweights small p ; underweights large p .
Inverse-S shape. $\gamma \approx 0.65$ (T&K 1992).

Four-Fold Pattern

- Low p , gain: overweight $\Rightarrow$ lottery
- Low p , loss: overweight $\Rightarrow$ insurance
- High p , gain: underweight $\Rightarrow$ take certain gain
- High p , loss: underweight $\Rightarrow$ gamble

Cumulative Prospect Theory (Tversky & Kahneman, 1992)

$$V(\text{prospect}) = \sum_i \pi_i \cdot v(x_i - r)$$

where r = reference point; $v(\cdot)$ = S-shaped value function ($\alpha = \beta = 0.88$, $\lambda = 2.25$); π_i = rank-dependent cumulative decision weights ($\gamma = 0.65$).

Why “Cumulative”?

Original 1979 PT violated first-order stochastic dominance. CPT (1992) fixes this by applying weights to *cumulative* probabilities — the rank-dependent approach.

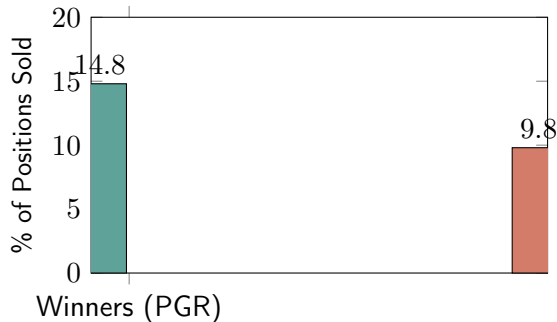
Empirical Fit

- Explains 90%+ of modal choices EU gets wrong (Camerer, 1995).
- Wu & Gonzalez (1996): fits well across cultures.
- Replicated in real money, high stakes, expert populations.

The Disposition Effect — PT in Action

Shefrin & Statman (1985); Odean (1998): Investors are more likely to **sell winners** (lock in gains) than **hold losers** (avoid realising a loss).

Disposition Effect: Odean (1998)



The Numbers

- PGR = **14.8%**; PLR = **9.8%**
- **52% more likely** to sell winners.
- Cost: $\approx 3.4\%$ lower annual returns.

Why PT predicts this: Concave (gain) region $\Rightarrow$ risk averse $\Rightarrow$ sell winners. Convex (loss) region $\Rightarrow$ risk seeking $\Rightarrow$ hold losers.

[GAME] Experience the Disposition Effect

You have two stocks in your portfolio. You need cash — sell one.

Stock A: You bought it at \$100. It is now worth \$130.

Stock B: You bought it at \$100. It is now worth \$70.

Which do you sell?

Show of Hands

How many chose Stock A (the winner)?

How many chose Stock B (the loser)?

What Finance Theory Says

Sell whichever has *worse future prospects*. Past purchase price is a **sunk cost** — irrelevant to future performance. Your decision should not depend on which one is a winner or loser vs your reference price.

What Prospect Theory Predicts

You sell Stock A (winner) because you are risk-averse in the gain domain and want to lock in the certain \$30 gain. You hold Stock B (loser) because you are risk-seeking in the loss domain — hoping for a recovery. This costs real money.

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Genesove & Mayer (2001) — Boston Condominiums

Homeowners with negative equity set **higher asking prices** and wait **longer to sell** than owners with positive equity. They refuse to realise a nominal loss.

Canadian Housing 2022–2023

After the Bank of Canada raised rates from 0.25% to 5.0%:

- Average home price fell 18% (peak Feb 2022 to Jan 2023).
- Sellers who bought near the 2022 peak: **avg 47 days on market** (vs 21 days for others).
- Many chose to **withdraw listings** rather than accept a nominal loss.
- MLS data shows bimodal listing distribution — clustered near purchase prices.

Policy Implication

Nominal loss aversion can slow market adjustment — creating price stickiness that prolongs housing downturns. Bank of Canada staff research (2023) documents this.

Rothman & Salovey (1997) — Message Framing and Health

Loss-framed vs gain-framed health messages have different effects depending on the behaviour:

- **Detection behaviours** (mammograms, skin cancer checks): **gain frame** works better. “Early detection saves lives.”
- **Prevention behaviours** (sunscreen, vaccines): **loss frame** works better. “Failing to use sunscreen means risking cancer.”

Organ Donation in Canada

- 80% of Canadians support organ donation in principle.
- Only 23% have formally registered (CBC, 2021).
- **Loss frame**: “Every year, 1,600 Canadians die waiting for an organ you could provide.”
- Nova Scotia opt-out legislation (2021): framing non-donation as *forgoing the opportunity to help* — registration rising.

Discussion Questions

- Q1. “Can you recall a purchase decision where you anchored on the purchase price and it affected your decision to sell or keep? How much did it cost you?”
- Q2. “Is loss aversion *irrational* or is it *evolutionarily adaptive*? What environments might it have been designed for?”
- Q3. “Knowing you are loss averse, can you overcome it? What strategies would help?”

Think–Pair–Share

Discuss Q2 with your neighbour. The evolutionary argument is actually quite deep — losses (starvation, predation) were historically more consequential than equivalent gains.

Key Takeaways — Lecture 3

- ✓ **Four-fold pattern:** risk averse (gains, high prob), risk seeking (losses, high prob), risk seeking (gains, low prob), risk averse (losses, low prob).
- ✓ **Reference dependence:** utility over *changes* from reference point, not final wealth.
- ✓ **Loss aversion:** $\lambda \approx 2.25$ — losses loom $2.25\times$ larger than equivalent gains.
- ✓ **Value function:** S-shaped — concave in gains, convex in losses, steeper for losses.
- ✓ **Probability weighting:** overweight small probs (lottery, insurance); underweight moderate-to-large.
- ✓ **Applications:** disposition effect, endowment effect, housing, health, wage rigidity.

Next: Lecture 4: Time Inconsistency & Present Bias — We examine a different dimension of irrationality: the temporal dimension. Why do we systematically make promises to ourselves that we break?

The Equity Premium Puzzle — Bonus

Benartzi & Thaler (1995) — Myopic Loss Aversion

US stocks have returned $\approx 6\%$ more per year than bonds for 130 years (Mehra & Prescott, 1985). Standard EU requires implausibly high risk aversion ($\gamma > 30$) to justify this.

The Behavioural Explanation

If investors:

1. Are loss averse ($\lambda = 2.25$)
2. Evaluate their portfolio **annually** (not over a lifetime horizon)

Then the required equity premium is *exactly* $\approx 6\%$.

Policy Implication

Less frequent portfolio statements (quarterly vs monthly) should increase stock market participation, because investors experience fewer “loss” periods. This is a testable, policy-relevant prediction from PT.

Further Reading

- Kahneman, D. & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291. **The original paper — essential.**
- Tversky, A. & Kahneman, D. (1992). Advances in prospect theory. *Journal of Risk and Uncertainty*, 5, 297–323.
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- Genesove, D. & Mayer, C. (2001). Loss aversion and seller behavior. *QJE*, 116(4), 1233–1260.
- Benartzi, S. & Thaler, R.H. (1995). Myopic loss aversion and the equity premium puzzle. *QJE*, 110(1), 73–92.